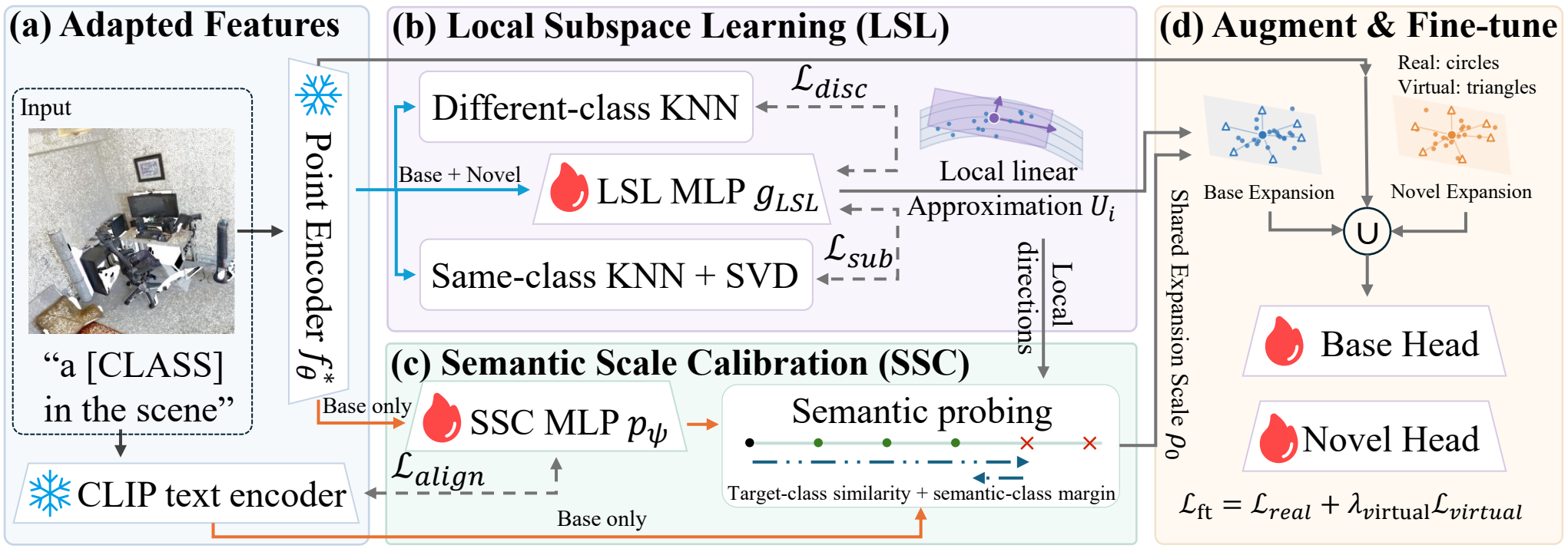

(a) Adapted Features
Input
"a [CLASS] in the scene"
CLIP text encoder
Point Encoder f_θ^*
(b) Local Subspace Learning (LSL)
Different-class KNN
$\mathcal{L}_{disc}$
Base + Novel
LSL MLP g_{LSL}
Local linear Approximation U_i
Same-class KNN + SVD
$\mathcal{L}_{sub}$
Local directions
(c) Semantic Scale Calibration (SSC)
Base only
SSC MLP p_ψ
$\mathcal{L}_{align}$
Semantic probing
Target-class similarity + semantic-class margin
Base only
(d) Augment & Fine-tune
Real: circles
Virtual: triangles
Base Expansion
Novel Expansion
U
Shared Expansion Scale ρ_0
Base Head
Novel Head
$\mathcal{L}_{\text{ft}} = \mathcal{L}_{real} + \lambda_{\text{virtual}}\mathcal{L}_{virtual}$